Strider Settgast

Physics & Computational Mathematics · Florida Institute of Technology

EDUCATION

Florida Institute of Technology — Melbourne, FL

B.S. in Physics, Minor in Computational Mathematics · Expected May 2027 · GPA 3.89 / 4.00

- Sigma Pi Sigma — National Physics Honor Society
- Relevant coursework: Quantum Mechanics, Electromagnetic Theory, Modern Physics, Complex Variables, Introduction to PDEs & Applications, Linear Algebra, Differential Equations, Differential Geometry (graduate), Deep Learning (graduate), Scientific Computing, Electronic Measurement & Instrumentation

RESEARCH

Undergraduate Researcher & Project Lead — Florida Institute of Technology

Advisor: Dr. Paul Martin · March 2026 – Present

- Developing an inverse beamforming method using spherical elementary current systems (SECS) to localize and characterize MHD wave sources from ground-based time-series magnetometer data.
- Derived the underlying mathematical framework; implemented the full analysis pipeline in Python (NumPy / SciPy / scikit-learn) from raw magnetometer time series through source reconstruction.
- Demonstrated, via a frequency-correlation spectrum across the full observation window, that spatially separated magnetometers detect coherent global periodic anomalies — evidence of a genuine large-scale signal rather than local noise.
- Lead a team of three researchers, coordinating workflow and mentoring members on methods and tooling.
- Preparing a first-author manuscript for submission (target: November 2026).

TECHNICAL SKILLS

Programming: Python (proficient), C++ (foundational), SQL, Bash

Scientific computing & ML: NumPy, SciPy, scikit-learn, PyTorch, MATLAB

Tools: Git, LaTeX

Methods: Signal processing, inverse problems, beamforming, time-series analysis, numerical modelling

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